

Making a Face Recogniser Work Under Different Lighting and Rotation

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October 6, 2026

1 What the recogniser does

A face photo is a grid of pixels, each with a brightness from black to white. For every pixel, the recogniser looks at the eight pixels around it and asks, is this neighbour brighter or darker than me? A brighter neighbour becomes a 1, a darker one a 0. The eight answers form a short code, like 10110010, which is a little signature of that spot: an edge, a line, or a flat area. The face is then divided into a grid of small windows, and in each window the recogniser counts how often every signature appears. All those counts together form a fingerprint of the face. To identify a person, the new fingerprint is compared with stored ones, and the closest match is selected. On ordinary photos, this works very well, recognising about 97.5% of the faces correctly.

2 The two problems

If one side of the face is much brighter than the other, the signatures change and the accuracy falls to about 70%. If the face is turned, the little circle of eight neighbours turns with it, the digits of each signature shift, and at a 20-degree turn, the accuracy drops to about 63%.

3 Fix 1: evening out the lighting

The key idea is to remove the lighting *before* the signatures are built, so the signatures never see the bright-to-dark glow in the first place. The lighting across a face is smooth and slowly changing, while the face itself has fine detail. So the recogniser makes a *blurry* copy of the photo, which keeps only the smooth glow and loses the fine detail, and then divides the original photo by that blurry copy. The smooth glow is the same in both, so it cancels out, while the fine face detail survives the division. What remains is the face, as if it had been lit perfectly evenly, and only then are the signatures and the fingerprint computed.

nose and mouth sharp but large enough to catch the lighting. Because this step only removes smooth brightness, it does nothing to a normal, evenly lit photo, so the accuracy on easy photos is not affected. This fix brings the strongly lit case back up to about 97.5%.

4 Fix 2: turning the faces while learning them

Here, the idea is to prepare for rotation when the faces are first stored. The recogniser not only keeps the straight-on photo of each known person; it also keeps several copies turned a little, one every five degrees from thirty degrees left to thirty degrees right. Each copy is just another

fingerprint in the database for the same person. When a turned photo then arrives, it does not have to match a straight-on template: it can match the turned copy that lines up with it, and so it is still recognised as the right person.

This works like keeping a few passport photos of a friend, one facing straight, one turned slightly left, and one turned slightly right. Whichever way the friend is looking, one of the photos still looks like them. Turning the stored faces is cheap, done once ahead of time, and it does not change the straight-on photos at all, so normal recognition is untouched. This fix brings the 20-degree case back up to about 98%.

5 The result

Table 1: Rank-1 recognition rate: baseline vs. improved.

Test condition	Baseline	Improved
Normal	0.975	0.975
Uniform brightness	0.975	0.975
Lighting ramp $s = 0.2$	0.960	0.975
Lighting ramp $s = 0.4$	0.945	0.970
Lighting ramp $s = 0.6$	0.895	0.985
Lighting ramp $s = 0.8$	0.705	0.975
Lighting ramp $s = 1.0$	0.335	0.905
Rotation 5°	0.970	0.985
Rotation 10°	0.925	0.990
Rotation 15°	0.800	0.980
Rotation 20°	0.630	0.980
Rotation 25°	0.510	0.960
Rotation 30°	0.340	0.965

The figure shows two terminal windows side-by-side. The left window shows the output of the 'make baseline' command, and the right window shows the output of the 'make improved' command. Both outputs are formatted as tables with test conditions and their corresponding recognition rates.

make baseline		make improved		
condition	baseline	condition	baseline	improved
normal	0.975	normal	0.975	0.975
uniform	0.975	uniform	0.975	0.975
lighting 0.2	0.960	lighting 0.2	0.960	0.975
lighting 0.4	0.945	lighting 0.4	0.945	0.970
lighting 0.6	0.895	lighting 0.6	0.895	0.985
lighting 0.8	0.705	lighting 0.8	0.705	0.975
lighting 1	0.335	lighting 1	0.335	0.905
rotation 5	0.970	rotation 5	0.970	0.985
rotation 10	0.925	rotation 10	0.925	0.990
rotation 15	0.800	rotation 15	0.800	0.980
rotation 20	0.630	rotation 20	0.630	0.980
rotation 25	0.510	rotation 25	0.510	0.960
rotation 30	0.340	rotation 30	0.340	0.965

Figure 1: Terminal output of make baseline (left) and make improved (right).